TASK 4 – TRAFFIC-ADAPTIVE TIMING

Adaptive Multi-Junction Traffic Grid Controller (AMTGC)

Project	Adaptive Multi-Junction Traffic Grid Controller (AMTGC)
Task	Task 4 – Traffic-Adaptive Timing
Simulation tool	Intel/Altera ModelSim

1. Objective

The objective of Task 4 is to enhance the AMTGC system by adjusting each junction's green-light duration according to its traffic-density input. The design uses a 3-bit traffic-density value for each junction and applies minimum and maximum timing limits to keep the adaptive timing within configured bounds.

2. Work Completed

• Added a 3-bit **traffic_density** input for Junction A and Junction B, representing density levels from 0 to 7.

• Updated **junction_controller.v** to calculate a green duration from the traffic-density input.

• Added minimum and maximum green-time limits to prevent the calculated duration from going below or above the configured bounds.

• Added a register to latch the calculated green duration at the beginning of a green phase, so a density change during that phase does not alter its active duration.

• Updated **amtgc_top.v** to pass the separate traffic-density inputs to the two junction controllers.

• Updated the testbenches to exercise different traffic-density combinations.

• Kept **generic_timer.v** unchanged; the controller supplies the timer with the calculated count target.

3. Adaptive Timing Calculation

The green-time calculation uses the configured minimum, the traffic-density level, and an extension amount per density level:

```
Green Time = MIN_GREEN_TIME + (traffic_density × GREEN_EXT_PER_LEVEL)
```

The result is then limited to the configured minimum and maximum values. In the current configuration, MIN_GREEN_TIME = 10, MAX_GREEN_TIME = 30, and GREEN_EXT_PER_LEVEL = 3 clock-count units per density level.

4. Files Used

File	Role
generic_timer.v	Programmable timer; unchanged from Task 3.
junction_controller.v	Updated controller with adaptive green-time calculation and limits.
ped_arbiter.v	Shared pedestrian request arbitration; carried forward from Task 3.
amtgc_top.v	Top-level integration of both junctions and their density inputs.

File	Role
tb_generic_timer.v	Timer testbench.
tb_junction_controller.v	Controller testbench updated for traffic density.
tb_amtgc_top.v	Top-level testbench with traffic-density combinations.

5. Compilation and Simulation

All seven Verilog files were successfully compiled in ModelSim after correcting a duplicate port connection in the top-level testbench. The top-level testbench was simulated and waveform signals were observed. The internal green_time_latched values were added to the waveform for inspection.

6. Observed Results

During the observed simulation, Junction A had traffic-density level 2 and its latched green time was 16 count units. Junction B had traffic-density level 7; its calculated value reached the configured maximum and was limited to 30 count units. These observations are consistent with the adaptive timing formula and maximum cap.

7. Verification Status and Limitations

Basic compilation and simulation were completed, and adaptive green-time behavior was observed for the tested values. A complete sweep of every density level from 0 through 7, as well as verification of green-wave coordination across multiple density combinations, has not yet been completed. These checks should be performed before claiming full verification.

8. Conclusion

The traffic-adaptive timing feature was implemented in the AMTGC design. Each junction receives its own traffic-density input, calculates a bounded green duration, and latches that duration for the active green phase. The design compiled successfully and basic simulation observations confirmed adaptive timing and maximum-time limiting. Further verification remains as described above.

Submission note: Attach the ModelSim waveform screenshot showing the density inputs and green_time_latched signals as simulation evidence, if required by your submission format.